

Final Assessment Runs using a4a

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Data input for the assessment

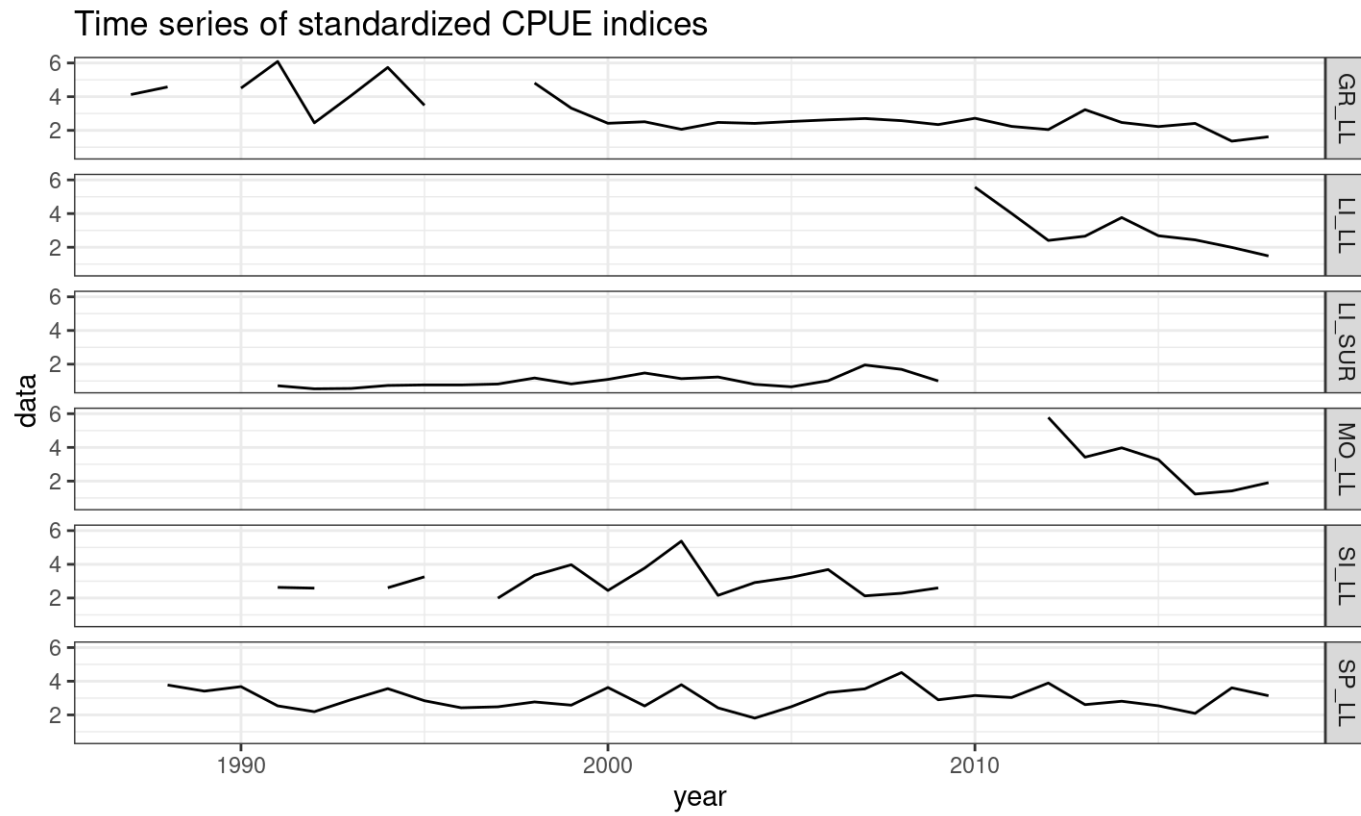
Catch Data

- Catch-at-age data since 1985
- plus group 5
- $\bar{f}$ at 2 - 4

CPUE Data

- Six standardized CPUE series
 - Greek longliners since 1987
 - Moroccan longliners since 2012
 - Spanish longliners since 1988
 - Ligurian longliners since 2010
 - Sicilian longliners from 1991 to 2009
 - Ligurian surface longliners 1991 to 2009
- Representative of age groups 2 - 4

CPUE indices



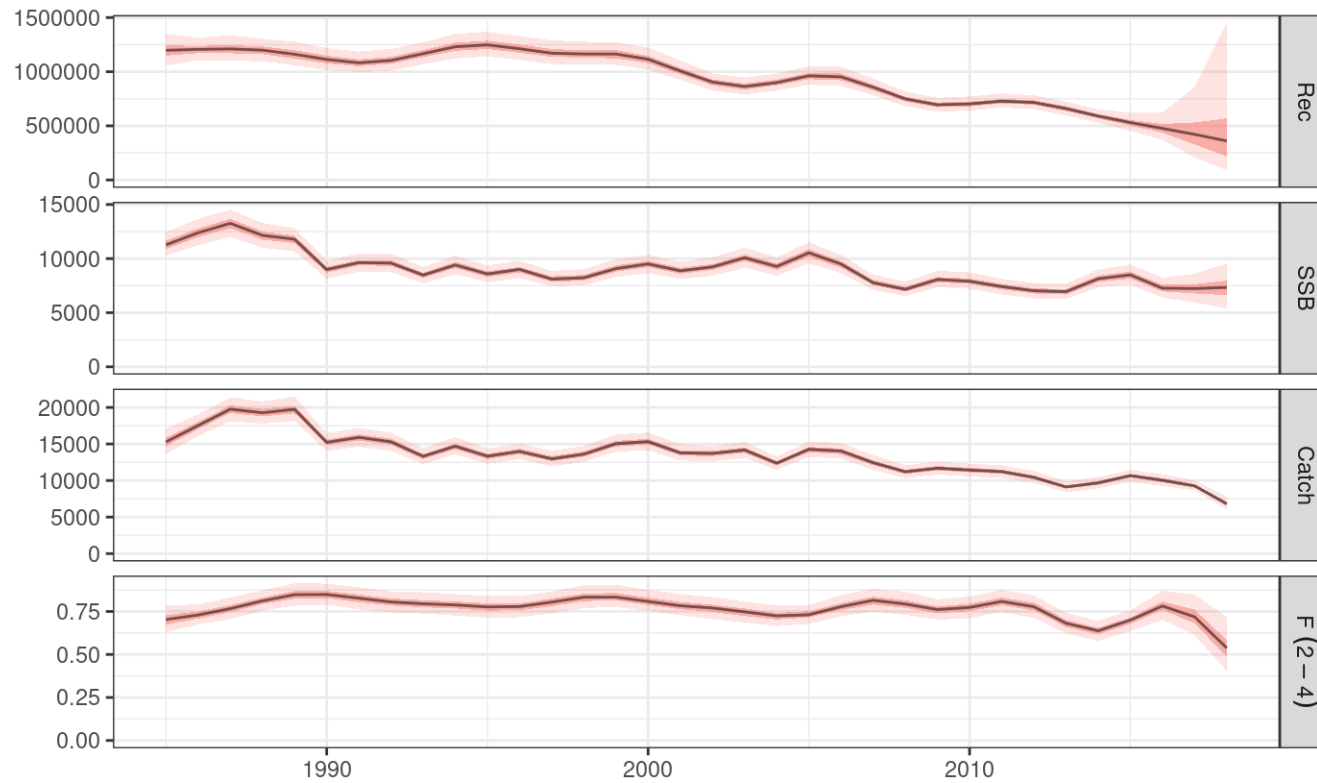
Final runs

- Two final runs were considered:
 - one with constant M
 - one with constant M including estimated Discards
- Models specifications kept the same:
 - for F, R, σ^2, τ_s^2 and N_1
 - Q model altered according to the new CPUE index:

$$Q : \begin{cases} \sim 1, & \text{Greek longline} \\ \sim 1, & \text{Spanish longline} \\ \sim s(\text{year}, k = 3), & \text{Moroccan longline} \\ \sim s(\text{year}, k = 3), & \text{Ligurian longline} \\ \sim 1, & \text{Sicilian longline} \\ \sim s(\text{year}, k = 9), & \text{Ligurian surface longline} \end{cases}$$

final run

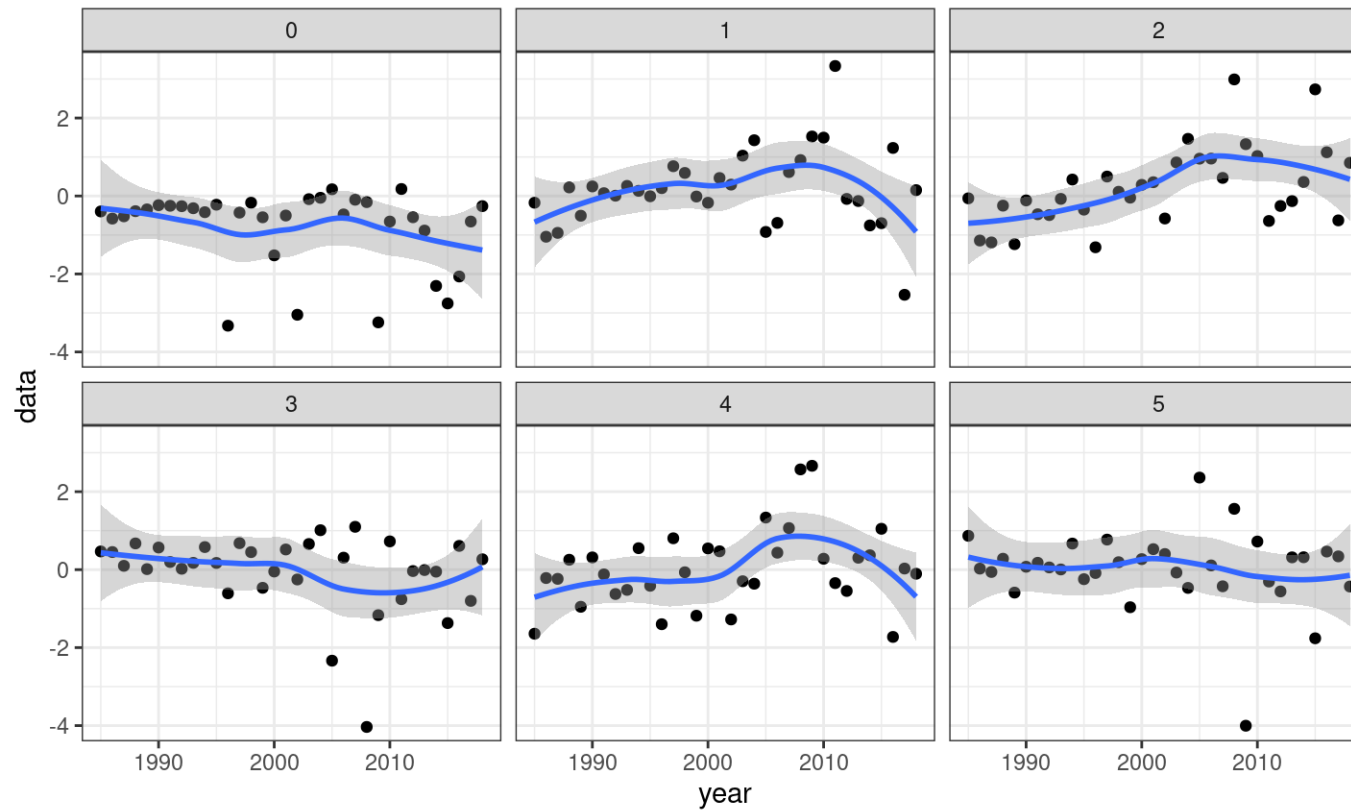
assuming constant M



Diagnostics

final run assuming constant M

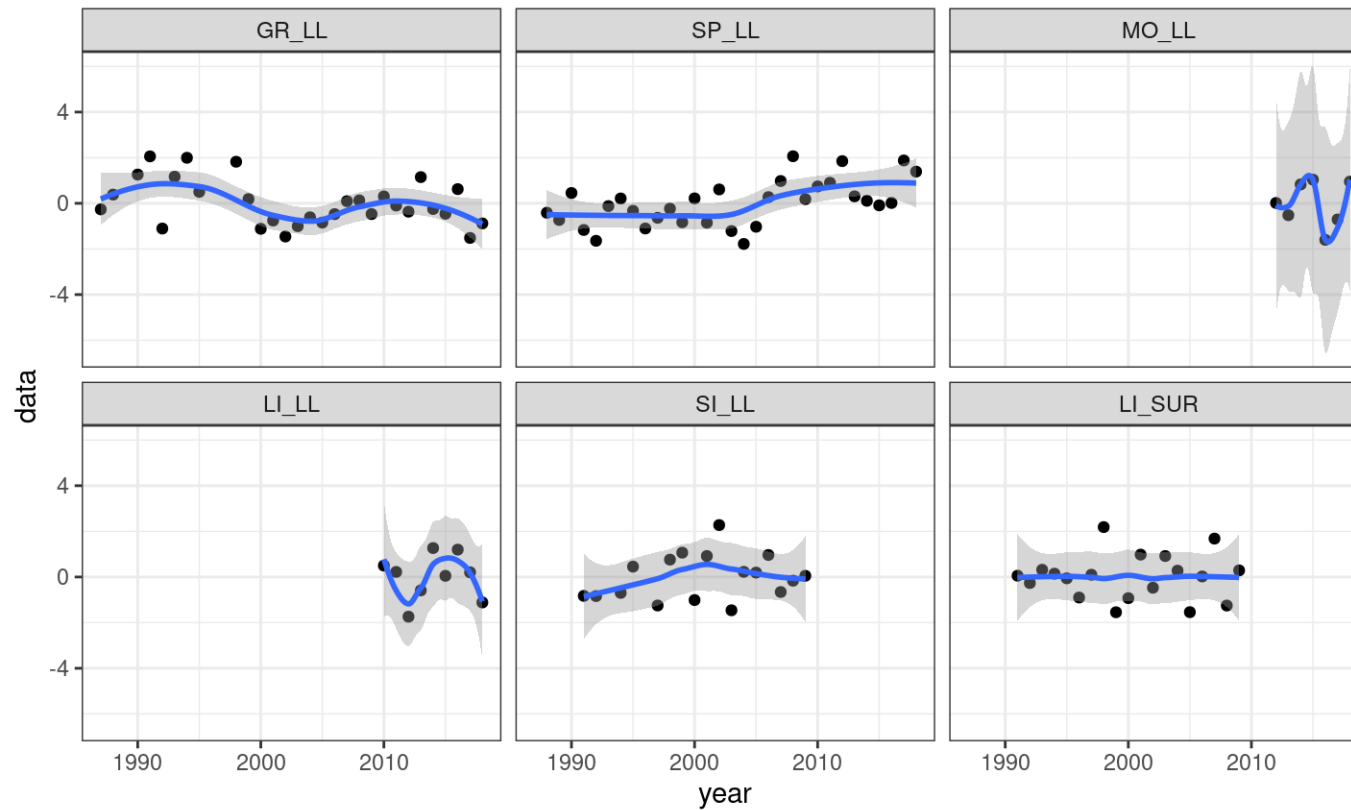
Standardized residuals of the catch numbers by age



Diagnostics

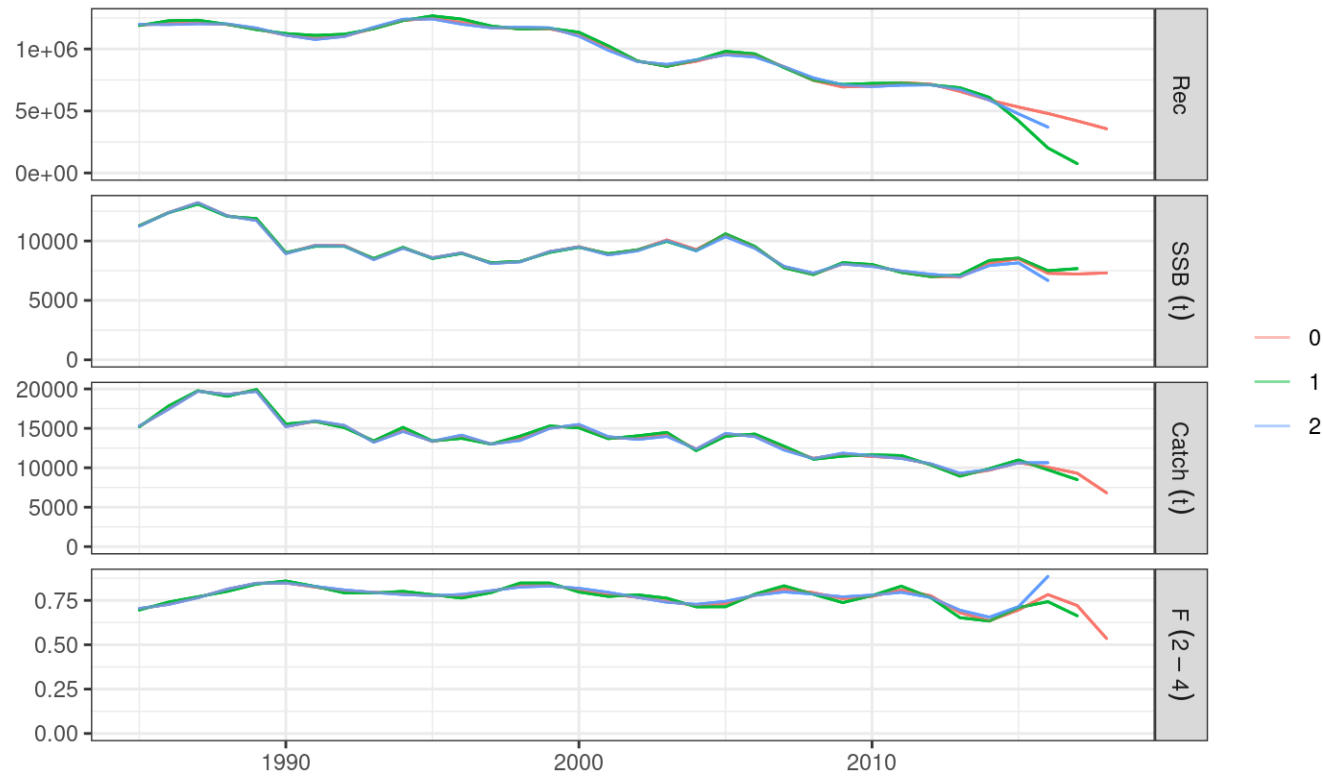
final run assuming constant M

Standardized residuals of the CPUE series



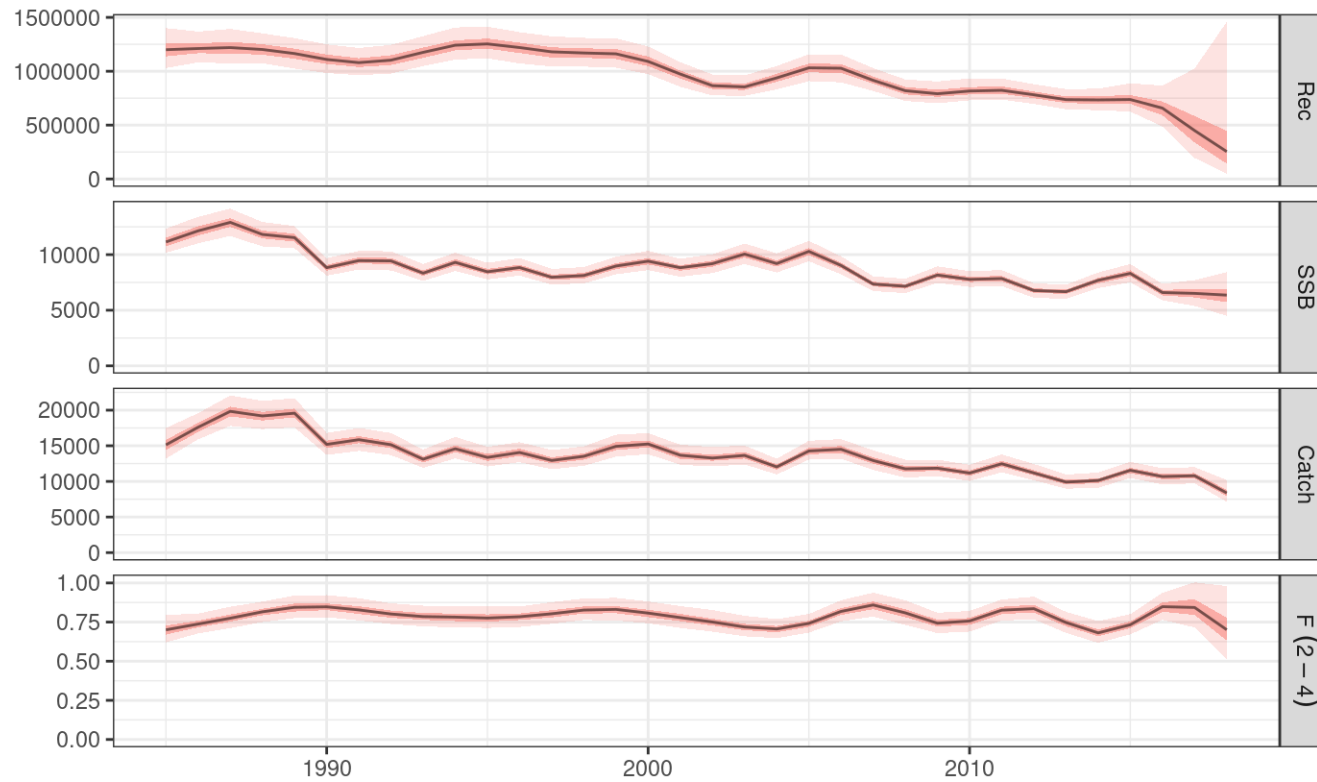
Retrospective plot

final run assuming constant M



final run

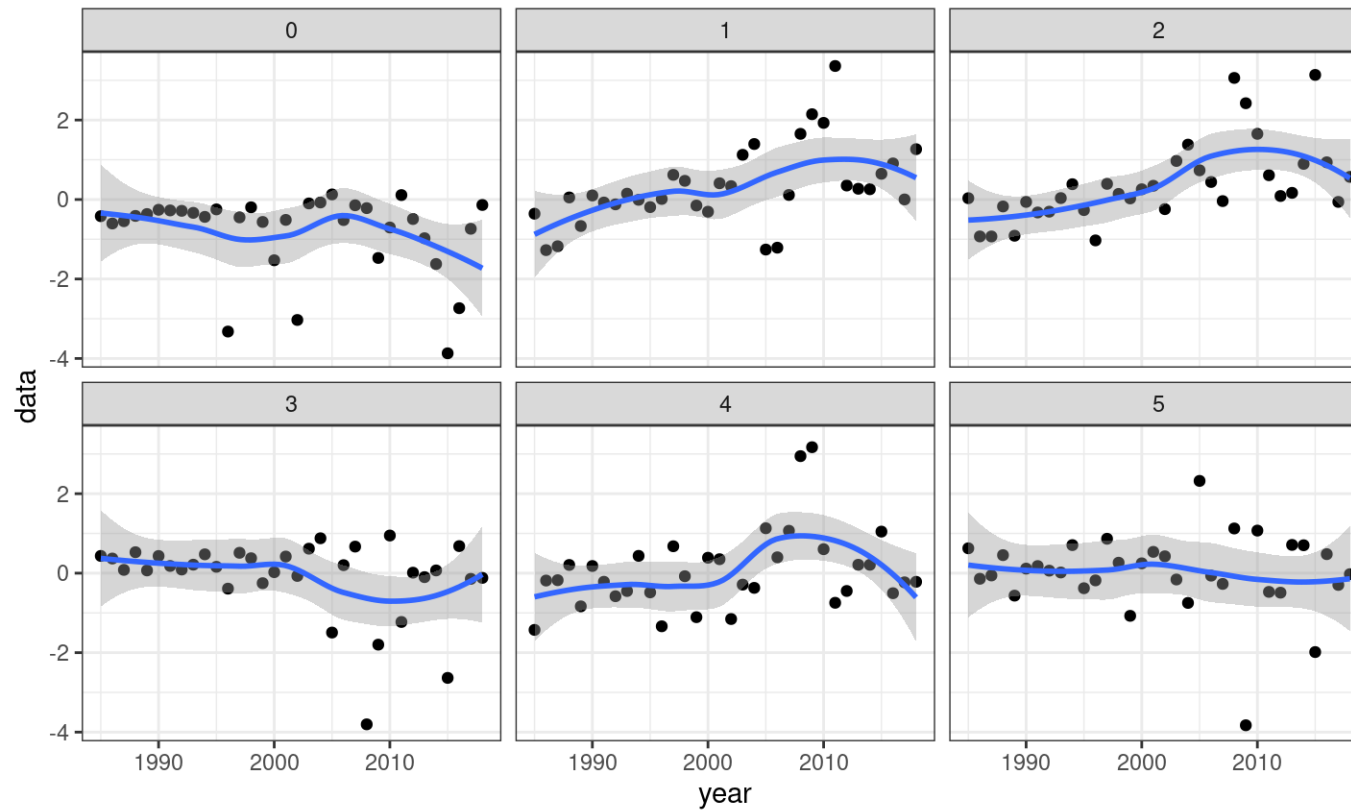
assuming constant M discards included



Diagnostics

final run assuming constant M discards included

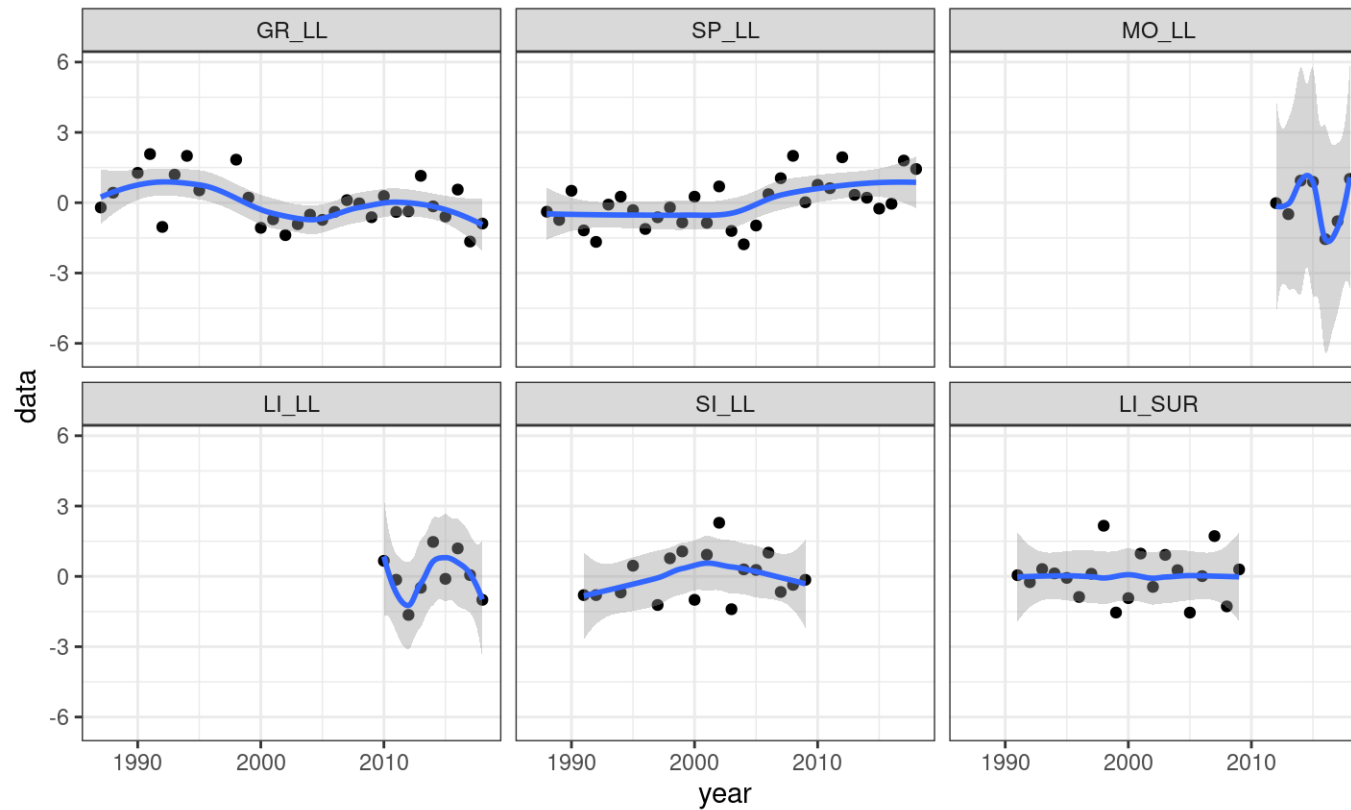
Standardized residuals of the catch numbers by age



Diagnostics

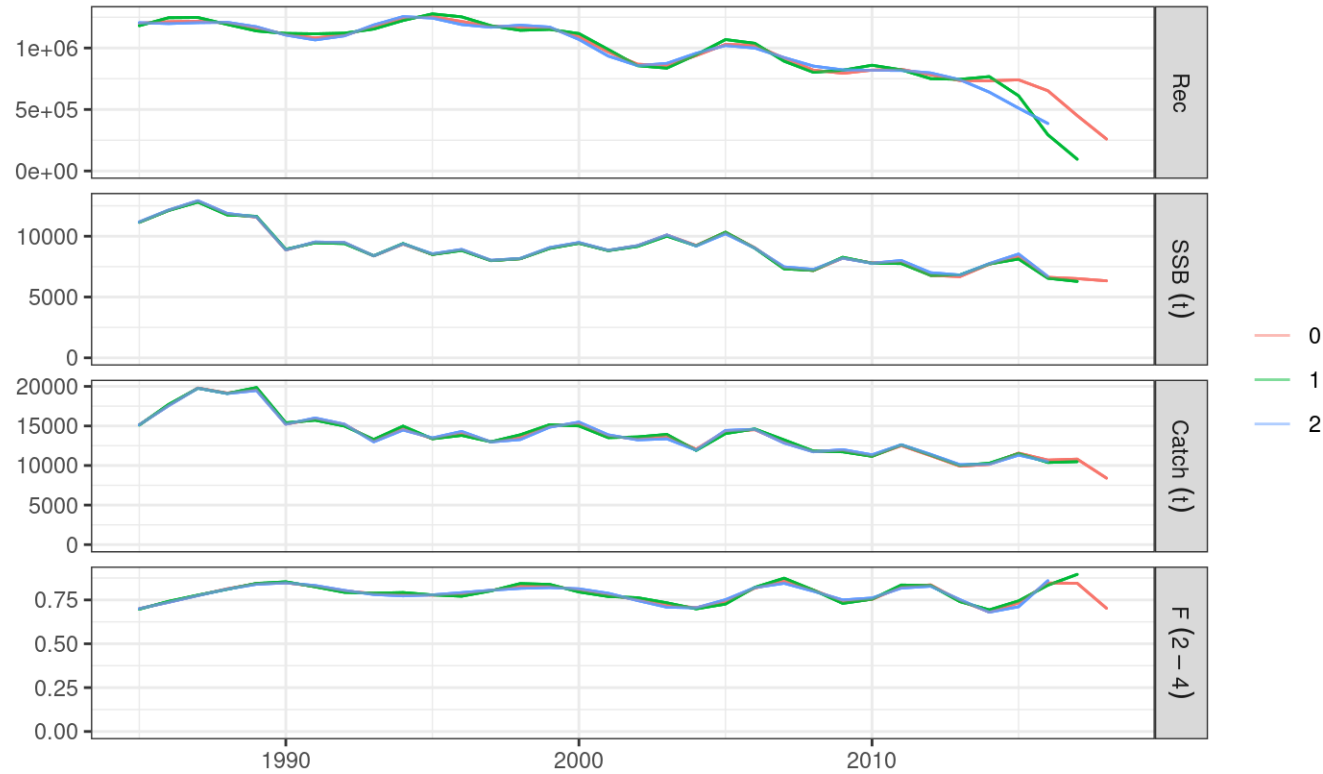
final run assuming constant M discards included

Standardized residuals of the CPUE series

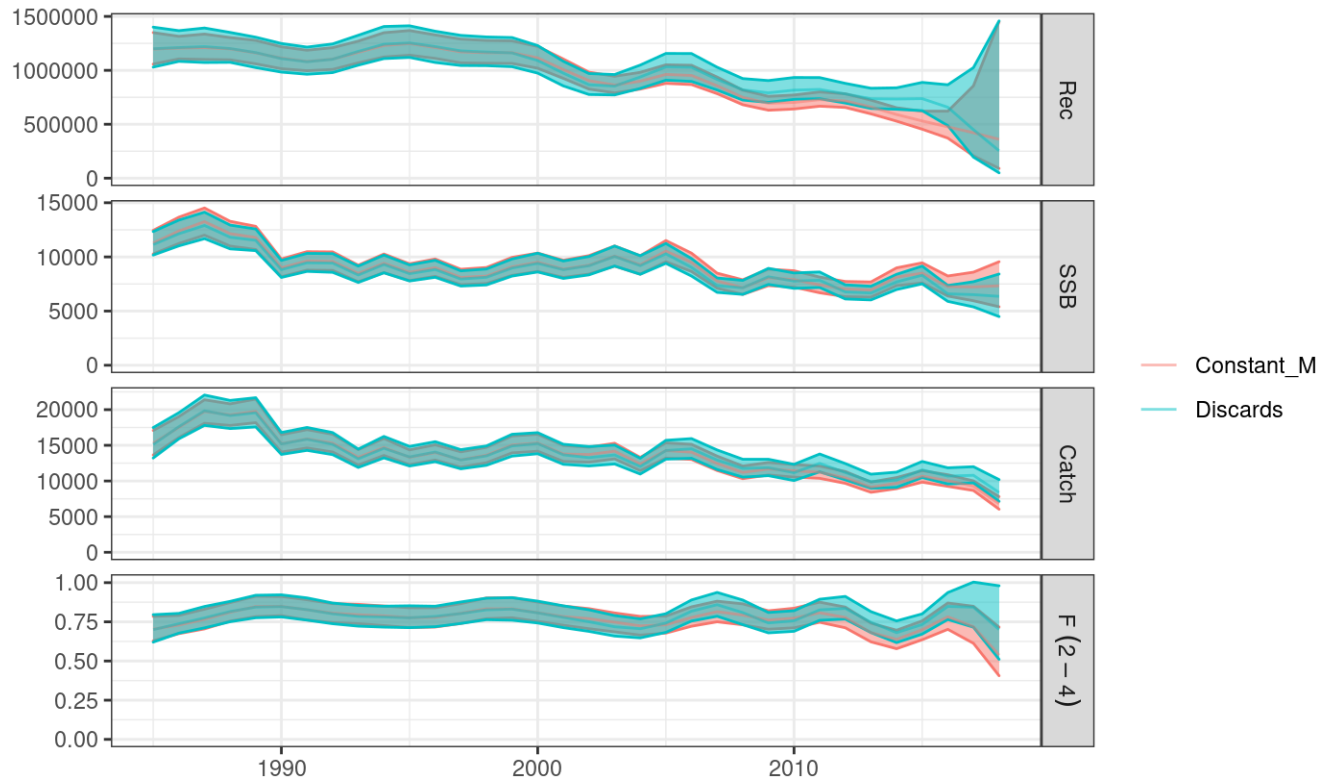


Retrospective plot

final run assuming constant M discards included



Comparison plot

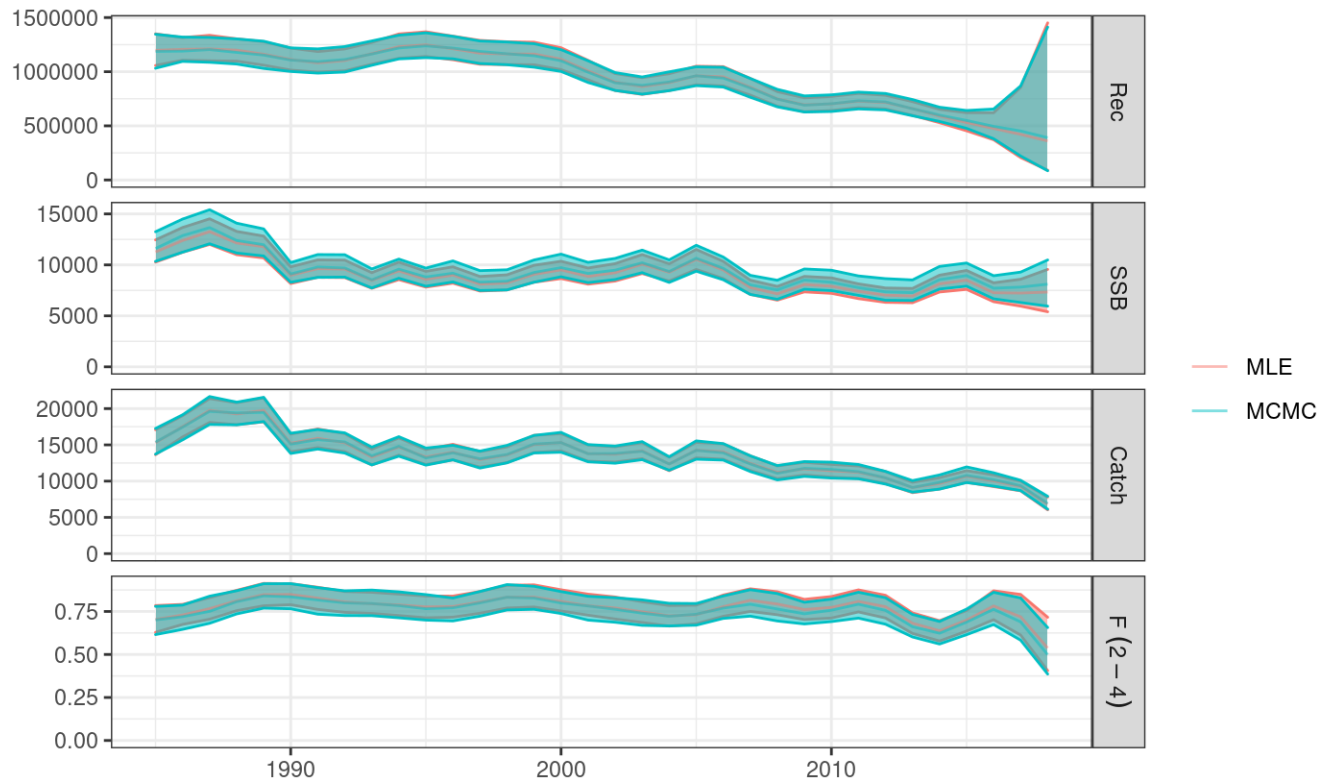


Reference points

value	constantM	Disc_constM
Fmsy	3.833524e-01	3.753627e-01
Bmsy	4.586190e+04	4.351330e+04
F2018	5.355717e-01	7.024563e-01
SSB	7.312521e+03	6.331603e+03
FFmsy	1.397074e+00	1.871407e+00
FF0.1	2.618664e+00	3.535289e+00
BBmsy	5.001516e-01	5.177004e-01

MCMC

constant M



MCMC

constant M discards included

